



Incidence of acute exacerbation of idiopathic pulmonary fibrosis in patients receiving antifibrotic agents: Real-world experience

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ABSTRACT

Background: Acute exacerbation of idiopathic pulmonary fibrosis (AE-IPF) is a fatal event that can occur during the clinical course of idiopathic pulmonary fibrosis (IPF). Although data from clinical trials suggest that the antifibrotic agents pirfenidone and nintedanib might reduce the risk of AE-IPF, the incidence of AE-IPF in patients receiving antifibrotic agents in clinical settings is unclear.

Objectives: To determine the incidence of AE-IPF in patients receiving antifibrotic agents and compare AE-IPF frequency in patients receiving pirfenidone and nintedanib.

Methods: We retrospectively reviewed the clinical records of 199 patients with IPF who were started on pirfenidone or nintedanib at our institution during the period from 2009 through 2018. Baseline characteristics, incidence of AE-IPF, and outcome after AE-IPF onset were analyzed.

Results: During the observation period, the 1-, 2-, and 3-year cumulative incidences of AE-IPF were 9.3 %, 22.1 %, and 25.0 %, respectively. The 1-, 2-, and 3-year cumulative incidence rates for AE-IPF in the pirfenidone group and nintedanib group were 5.1 % vs. 18.6 %, 20.4 % vs. 25.2 %, and 22.6 % vs. 29.6 %, respectively. AE-IPF incidence was significantly lower in patients treated with pirfenidone than in those treated with nintedanib (log rank test, $P = 0.035$). The 3-month survival rate after AE-IPF onset was 61.1 % in the pirfenidone group and 61.5 % in the nintedanib group; thus, outcomes after AE-IPF onset were similar in the 2 groups.

Conclusion: The reduction in AE-IPF risk might be greater for pirfenidone than for nintedanib.

1. Introduction

Idiopathic pulmonary fibrosis (IPF) is a lung disorder characterized by fibrosis of the lung interstitium and is the most common subtype of chronic fibrosing interstitial pneumonia [1]. Although the clinical course of IPF is usually chronic and progressive, acute exacerbation of IPF (AE-IPF) develops in some IPF patients. AE-IPF is diagnosed clinically when a patient with a previous or existing diagnosis of IPF develops acute respiratory worsening, typically within 1 month, in the absence of other known causes such as heart failure or fluid overload [2]. AE-IPF is characterized radiologically by new ground-glass opacities in bilateral lung fields, superimposed on existing reticular shadows. Histopathologically, AE-IPF manifests as diffuse alveolar damage superimposed on pulmonary fibrosis with a usual interstitial pneumonia pattern. Although a previous guideline in Japan defined AE-IPF as unexplained worsening or development of IPF, some patients develop AE-IPF after

direct or indirect lung injury [3]. New diagnostic criteria proposed by the International Working Group in 2016 divide AE-IPF into idiopathic and triggered subtypes [2]. Triggers include infection, drug toxicity, aspiration, and invasive procedures such as surgery. The prognosis of AE-IPF is poor: median survival after incident AE-IPF is 3–4 months [4, 5]. In addition, a Japanese epidemiological survey found that AE-IPF was a leading cause of death in IPF patients and that about 40 % patients died from AE-IPF [6]. Therefore, prevention of AE-IPF would be a clinically important improvement for IPF treatment.

To date, 2 antifibrotic agents—pirfenidone and nintedanib—are available for treatment of IPF. Pirfenidone suppresses inflammatory cytokines and growth factors; however, its direct targets are unknown [7]. Nintedanib is a triple-receptor tyrosine kinase inhibitor that inhibits fibroblast growth factor, platelet-derived growth factor, and vascular endothelial growth factor [8]. Both drugs suppressed chronic progression of IPF [9–14], and clinical studies of antifibrotic agents suggest that

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they may help protect against AE-IPF [9,13–16]. In a phase 2 clinical trial in Japan, none of the patients treated with pirfenidone developed AE-IPF during a 9-month period, but 14 % of patients in a placebo group developed AE-IPF [9]. The efficacy of nintedanib for IPF patients was shown in 2 large replicate phase 3 clinical trials: INPULSIS 1 and 2. In INPULSIS-2 [14], AE-IPF incidence was significantly lower in a nintedanib-treated group than in a placebo group. Although there was no significant difference in INPULSIS-1, pooled analysis showed a lower investigator-reported incidence in the nintedanib treatment group than in the placebo group [16].

Existing evidence suggests that these antifibrotic agents might help prevent AE-IPF development. However, the incidence of AE-IPF in patients receiving fibrotic agents in clinical settings is unknown. In addition, to our knowledge no studies have directly compared the incidence of AE-IPF among patients receiving pirfenidone and those receiving nintedanib in a single center. The present retrospective analysis determined AE-IPF incidence and examined the clinical characteristics of patients receiving antifibrotic agents.

2. Methods

2.1. Patients

We reviewed the medical records of consecutive patients started on the antifibrotic agents pirfenidone or nintedanib for IPF at our institution during the period from 2009 through 2018. Data on patient characteristics, development of AE-IPF, treatment, and outcome after AE-IPF were collected from medical records. Data on the following patient characteristics at the start of antifibrotic treatment with either drug were collected: sex, smoking history, laboratory data, pulmonary function test results, and Gender-Age-Physiology (GAP) score. IPF was diagnosed by using the guidelines of the American Thoracic Society/European Respiratory Society/Japanese Respiratory Society/Latin American Thoracic Association [1]. Diagnosis of acute exacerbation of IPF was based on criteria reported by the International Working Group of Acute Exacerbation of IPF [2]. Patients concomitantly treated with both agents were excluded from the study. Patients who switched from a study drug to another antifibrotic agent because of an adverse event or inadequate response during the observation period were included. In addition, because we aimed to analyze AE-IPF incidence during the natural course of the disease in patients receiving antifibrotic agents, we excluded patients who developed triggered AE-IPF. The interval from the start of antifibrotic therapy to the first AE-IPF event was evaluated. Patient characteristics such as P/F ratio, treatment for AE-IPF, and outcome after AE-IPF onset were also collected from medical records. The data were locked at the end of 2020. The institutional review board of Toho University Omori Medical Center approved this study (M19223). The requirement of patient consent was waived because this study was retrospective and noninvasive, and because patient anonymity was secured.

2.2. Data analysis

Continuous variables were compared with the unpaired *t*-test and Mann–Whitney *U* test, as appropriate. Categorical variables were compared using the χ^2 and Fisher exact test. We used the Kaplan–Meier method to detect cumulative development of AE-IPF and outcome after onset of AE-IPF. The log-rank test was used to compare 2 groups. Hazard ratios along and 95 % confidence intervals were calculated by using Cox proportional hazards analysis to identify independent predictors of AE-IPF development. All *P* values were two-sided and a *P* value of less than 0.05 was considered to indicate statistical significance. Statistical analysis was performed with SPSS version 27 (SPSS Inc., IL, USA).

3. Results

3.1. Patient characteristics

We identified 199 patients treated with pirfenidone or nintedanib. Two patients were treated with both pirfenidone and nintedanib and were excluded. In addition, 2 patients who developed triggered AE-IPF, 1 IPF patient with lung cancer who developed AE-IPF after chemotherapy, and 1 patient who developed AE-IPF after definite aspiration were classified as having triggered AE-IPF and excluded from the study. Ultimately, data from 195 patients were included in the analysis.

The patients were classified into 2 groups, namely, a pirfenidone group and nintedanib group. The characteristics of the patients are summarized in Table 1. During the observation period, 131 patients received pirfenidone and 64 patients received nintedanib. Because pirfenidone was approved earlier than nintedanib for treatment of IPF in Japan, more patients received pirfenidone, the mean administration period was longer, and few patients were treated with another antifibrotic agent before pirfenidone treatment. There was no significant difference in age, sex, smoking history, presence of chronic obstructive pulmonary disease, or past AE history between the 2 groups. Although serum surfactant protein-D (SP-D) level was significantly higher in the pirfenidone group, other biomarkers, pulmonary function, and severity stage were similar between groups. Because we previously administered *N*-acetylcysteine (NAC) therapy for treatment of IPF, inhaled NAC was much more frequently used in combination with pirfenidone.

3.2. Incidence of AE-IPF

Overall, 15.9 % of IPF patients receiving antifibrotic agents developed AE-IPF during the entire follow-up period. The 1-, 2-, and 3-year cumulative incidence rates of AE-IPF from the first visit to our center were 1.6 %, 7.1 %, and 10.1 %, respectively. As shown in Fig. 1, the 1-,

Table 1
Baseline characteristics of patients treated with antifibrotic agents.

	Overall	Pirfenidone	Nintedanib	P value
No.	195	131	64	–
Duration of treatment, days	653 ± 633	732 ± 696	507 ± 438	0.018
Past use of antifibrotic agent, n	19 (10 %)	2 (2 %)	17 (27 %)	<0.001
Age	74 ± 8	74 ± 7	73 ± 9	0.576
Male, n	157 (81 %)	102 (78 %)	55 (86 %)	0.248
History of smoking, n	156 (80 %)	105 (81 %)	51 (80 %)	0.860
COPD, n	40 (21 %)	26 (20 %)	14 (22 %)	0.850
GAP stage (1/2/3)	47/108/40	33/72/26	14/36/14	0.646
Laboratory data				
LDH, IU/L	266 ± 83	264 ± 84	272 ± 81	0.827
SP-D, ng/dL	312 ± 237	337 ± 258	260 ± 178	0.034
KL-6, U/mL	1187 ± 1107	1173 ± 1208	1210 ± 870	0.545
Pulmonary function test				
FVC, mL	1924 ± 735	1931 ± 680	1910 ± 843	0.850
%FVC, %	64.5 ± 21.6	65.6 ± 19.7	62.2 ± 25.0	0.301
FEV1.0 %, %	83.6 ± 18.4	85.1 ± 14.0	80.4 ± 24.9	0.092
%DLco, %	45.4 ± 22.1	45.8 ± 19.7	44.6 ± 26.3	0.715
Previous AE-IPF, n	26 (13 %)	19 (14 %)	7 (11 %)	0.655
Combination therapy				
Prednisolone, n	101 (52 %)	70 (53 %)	31 (48 %)	0.512
Inhaled NAC, n	40 (21 %)	38 (29 %)	2 (3 %)	<0.001

Data are presented as mean ± SD. COPD, chronic obstructive pulmonary disease; GAP, gender-age-physiology; LDH, lactate dehydrogenase; SP-D, surfactant protein-D; KL-6, Krebs von den Lungen-6; FVC, forced vital capacity; FEV: forced expiratory volume in 1 s; DLco, diffusing capacity for carbon monoxide; AE-IPF: acute exacerbation of idiopathic pulmonary fibrosis NAC: *N*-acetylcysteine.

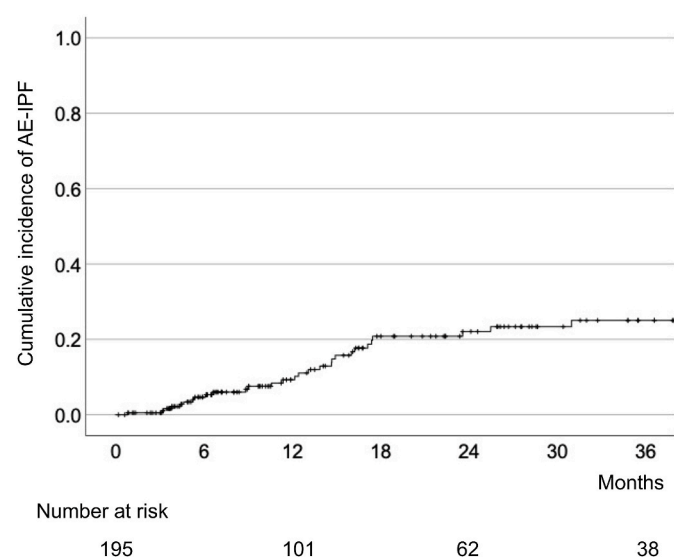


Fig. 1. Kaplan–Meier curve of accumulated incidence of acute exacerbation of idiopathic pulmonary fibrosis (AE-IPF) in all patients ($n = 195$). The estimated incidences were 9.3 %, 22.1 %, and 25.0 %, respectively at 1, 2, and 3 years.

2-, and 3-year cumulative incidence rates after the introduction of either antifibrotic agent were 9.3 %, 22.1 %, and 25.0 %, respectively. The cumulative incidence rate of AE-IPF was significantly lower in patients treated with pirfenidone than in those treated with nintedanib ($P = 0.035$) (Fig. 2). The 1-, 2-, and 3-year cumulative incidence rates for AE-IPF in the pirfenidone group and nintedanib group were 5.1 % vs. 18.6 %, 20.4 % vs. 25.2 %, and 22.6 % vs. 29.6 %, respectively. In a sub-analysis of patients who started antifibrotic agents as first-line therapy, the Kaplan–Meier curve was similar (Fig. 3A). Although it did not reach statistical significance in the whole-time analysis ($P = 0.136$), the 1-year incidence rate of AE-IPF was significantly lower in the pirfenidone group than in the nintedanib group (4.0 % vs. 16.4 %, $P = 0.008$).

In the pirfenidone group, 29 % of patients had received inhaled NAC. Thus, to exclude the effect of inhaled NAC therapy against AE-IPF development, we analyzed AE-IPF incidence in patients who had not

used inhaled NAC. As shown in Fig. 3B, even after limiting the analysis to patients not receiving NAC therapy, AE-IPF incidence was lower for patients treated with pirfenidone ($n = 93$) than for those treated with nintedanib ($n = 62$) ($P = 0.040$).

3.3. Characteristics of patients who developed AE-IPF and variables predicting development of AE-IPF

To identify other factors associated with AE-IPF development, we classified patients according to AE-IPF occurrence. Table 2 shows the baseline characteristics of the patients. There was no significant difference between groups in age, sex, smoking history, IPF severity, serological markers, pulmonary function, past AE history, or combination treatment with corticosteroids at baseline. We used Cox proportional hazards analysis to identify predictors of AE-IPF in patients treated with antifibrotic agents. As shown in Table 3, the antifibrotic drug selected was the only independent variable identified in multivariate analysis (HR 2.979 [95 % CI: 1.151, 6.332], $P = 0.014$).

3.4. Clinical variables and AE-IPF development and outcomes

Clinical characteristics at and after AE-IPF onset in pirfenidone-treated patients ($n = 18$) and nintedanib-treated patients ($n = 13$) are shown in Table 4. There was no significant difference at AE-IPF onset in $\text{PaO}_2/\text{FiO}_2$ ratio, serological markers, or treatment for AE-IPF between groups. All patients continued antifibrotic treatment after AE-IPF onset until either discharge from hospital because of disease improvement or death. We analyzed outcomes after AE-IPF onset (Fig. 4). The survival rate after AE-IPF onset was similar: the 3-month survival rate was 61.1 % in the pirfenidone group and 61.5 % in the nintedanib group. Overall median survival after AE-IPF onset was 267 days. Finally, we analyzed outcomes after the introduction of each drug (Fig. 5). The overall survival for patients receiving antifibrotic agents was similar in both groups (log rank test, $P = 0.634$).

4. Discussion

This study illustrates the real-world experience of AE-IPF in patients receiving antifibrotic agents. The annual incidence of AE-IPF was 5 %–15 % among IPF patients in cohort studies [4,17]. Among the 195 present patients, 31 (15.9 %) developed AE-IPF during the observation period. The 1-, 2-, and 3-year incidence rates were 9.3 %, 22.1 %, and 25.0 %, respectively, and these rates are consistent with those reported in other observational studies. Although there are no formal recommendations for preventing AE-IPF [18], clinical trials suggest that AE-IPF incidence might have decreased after introduction of antifibrotic agents [9,13,14]. In a phase 2 clinical trial of pirfenidone in Japan, none of the patients in a pirfenidone group developed AE-IPF, while 14 % of the patients developed AE-IPF in a placebo group during a 9-month period [9]. However, in a phase 3 clinical trial in Japan, a pirfenidone group and placebo group had similar incidence rates after a 52-week treatment period [10]. Incidence of AE-IPF was not included as an end point in the CAPACITY and ASCEND international phase 3 clinical trials of pirfenidone [11,12]. In a pooled analysis of these studies, however, the risk of respiratory hospitalization was lower for the pirfenidone arm than for the placebo arms [19]. Although the hospitalized patients might have included AE-IPF patients, the preventive effect was unclear, as details on the cause of respiratory-related hospitalization were not included. Interestingly, pirfenidone may prevent postoperative AE-IPF. Among 32 Japanese patients with IPF and lung carcinoma who received pirfenidone at least 2 weeks before lung resection [20], only 1 developed triggered AE-IPF within 30 days after the operation.

In the TOMORROW trial—a phase 2 study of nintedanib—the nintedanib group, especially those receiving 300 mg, had a lower incidence of AE-IPF per 12 months than did a placebo group [13]. In the INPULSIS trial, 2 pivotal phase 3 clinical trials both showed reduced forced vital

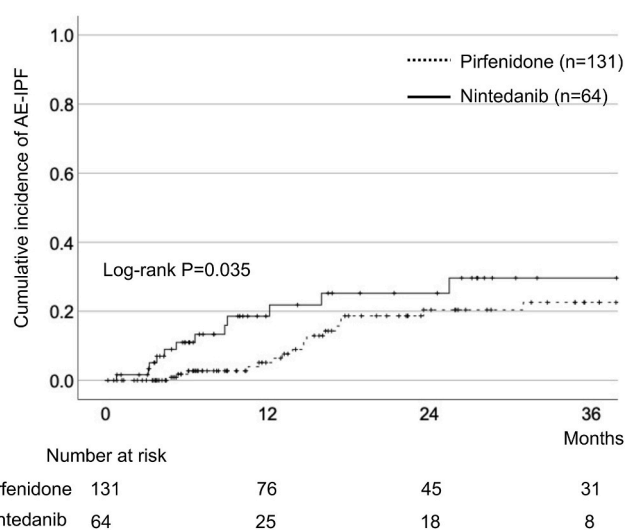


Fig. 2. Kaplan–Meier curve of accumulated incidence of acute exacerbation of idiopathic pulmonary fibrosis (AE-IPF) in patients receiving pirfenidone ($n = 131$) and nintedanib ($n = 64$). Cumulative incidence was significantly lower in the pirfenidone group ($P = 0.035$). The 1-, 2-, and 3-year cumulative AE-IPF incidence rates for the pirfenidone and nintedanib group were 5.1 % vs. 18.6 %, 20.4 % vs. 25.2 %, and 22.6 % vs. 29.6 %, respectively.

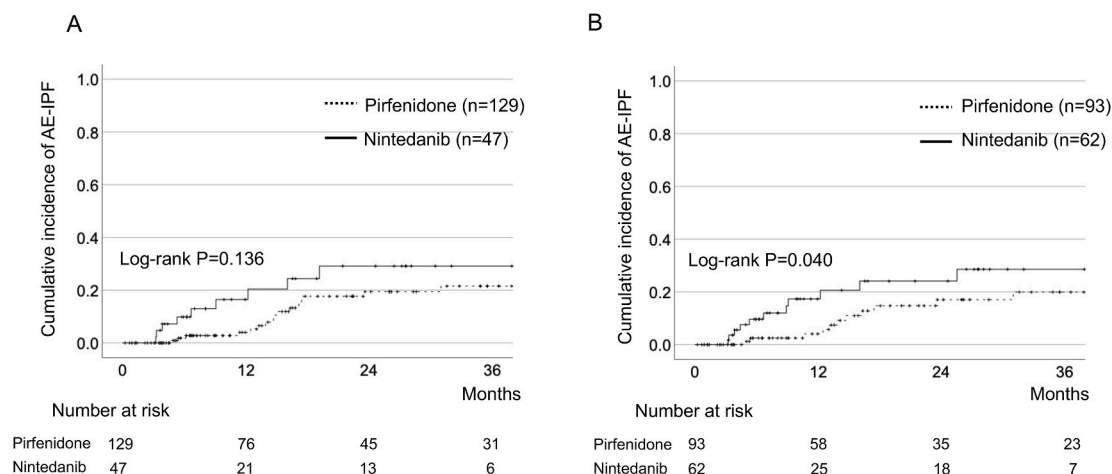


Fig. 3. (A) Accumulated acute exacerbation of idiopathic pulmonary fibrosis (AE-IPF) incidence rates for patients receiving first-line antifibrotic treatment. The 1-, 2-, and 3-year cumulative incidence rates for AE-IPF in the pirfenidone group ($n = 129$) and nintedanib group ($n = 47$) were 4.0 % vs. 16.4 % ($P = 0.008$), 19.7 % vs. 29.1 % ($P = 0.101$), and 21.6 % vs. 29.1 % ($P = 0.110$), respectively. (B) After excluding patients treated with inhaled *N*-acetylcysteine (NAC), Kaplan–Meier analysis showed a significantly lower risk of AE-IPF in the pirfenidone group ($n = 93$) than in the nintedanib group ($n = 62$) ($P = 0.040$).

Table 2

Baseline characteristics of IPF patients with and without AE-IPF during the study period.

	AE-IPF (–)	AE-IPF (+)	P value
No.	164	31	
Age	74 ± 8	73 ± 7	0.360
Male, n (%)	131 (80 %)	26 (84 %)	0.265
History of smoking, yes, n	37 (23 %)	3 (10 %)	0.103
COPD, n	34 (21 %)	6 (19 %)	0.742
GAP stage (1/2/3)	39/88/37	5/18/8	0.567
Laboratory data			
LDH, IU/L	266 ± 86	267 ± 70	0.910
SP-D, ng/dL	311 ± 252	303 ± 138	0.775
KL-6, U/mL	1184 ± 1158	1236 ± 940	0.778
Pulmonary function test			
FVC, mL	1931 ± 736	1847 ± 724	0.641
%FVC, %	65.0 ± 21.6	62.2 ± 20.5	0.475
FEV1.0 %, %	83.5 ± 18.7	84.4 ± 16.2	0.778
%DLco, %	46.3 ± 22.4	42.8 ± 20.7	0.380
Previous AE-IPF, n	20 (12 %)	4 (13 %)	0.912
Combination therapy			
Prednisolone, n	82 (50 %)	19 (61 %)	0.249
Inhaled NAC, n	33 (20 %)	7 (23 %)	0.756

Data are presented as mean ± SD. COPD, chronic obstructive pulmonary disease; GAP, gender-age-physiology; LDH, lactate dehydrogenase; SP-D, surfactant protein-D; KL-6, Krebs von den Lungen-6; FVC, forced vital capacity; FEV, forced expiratory volume in 1 s; DLco, diffusing capacity for carbon monoxide; AE-IPF, acute exacerbation of idiopathic pulmonary fibrosis; NAC, *N*-acetylcysteine.

Table 3

Cox proportional hazards analysis of AE-IPF risk in patients treated with anti-fibrotic agents.

	Hazard ratio	95%CI	P value
Age	1.002	0.956–1.051	0.956
Male	1.162	0.434–3.114	0.765
Nintedanib treatment ^a	2.937	1.202–7.179	0.018
Past use of antifibrotic agent	0.799	0.230–2.774	0.723
%FVC	0.989	0.975–1.004	0.149
SP-D	0.139	0.999–1.002	0.425
Combination treatment with NAC	1.989	0.752–5.260	0.166

AE-IPF, acute exacerbation of idiopathic pulmonary fibrosis; CI, confidence intervals; FVC, forced vital capacity; SP-D, surfactant protein-D; NAC, *N*-acetylcysteine.

^a Versus pirfenidone treatment.

Table 4

Clinical characteristics at onset of AE-IPF in patients receiving antifibrotic agents.

	Pirfenidone	Nintedanib	P value
No.	18	13	–
PaO ₂ /FiO ₂ ratio	284 ± 82	264 ± 67	0.463
Laboratory data			
LDH, IU/L	335 ± 81	331 ± 79	0.885
SP-D, ng/dL	380 ± 266	346 ± 161	0.668
KL-6, U/mL	1101 ± 478	1773 ± 1510	0.144
CRP, mg/dL	5.7 ± 4.4	7.1 ± 6.8	0.531
Treatment for AE-IPF			
Intravenous high-dose corticosteroids, n	18 (100 %)	13 (100 %)	–
rhTM, n	6 (35 %)	6 (46 %)	0.547
Antibiotics, n	13 (77 %)	12 (92 %)	0.249

Data are presented as mean ± SD. LDH, lactate dehydrogenase; SP-D, surfactant protein-D; KL-6, Krebs von den Lungen-6; AE-IPF, acute exacerbation of idiopathic pulmonary fibrosis; rhTM, recombinant human soluble thrombomodulin.

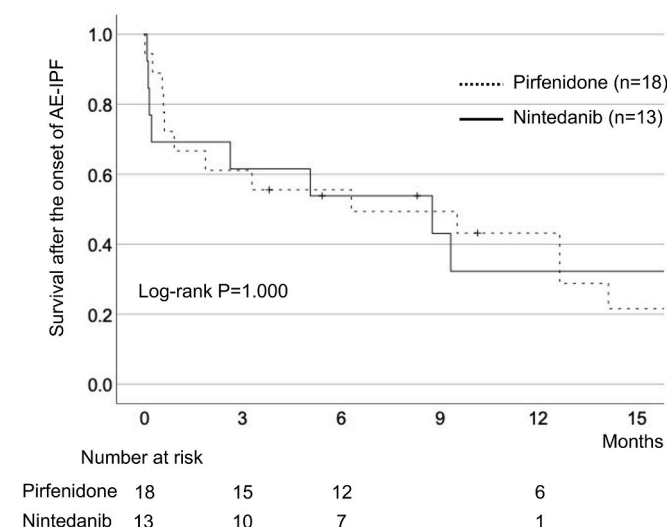


Fig. 4. Survival curve after onset of acute exacerbation of idiopathic pulmonary fibrosis (AE-IPF) in patients treated with pirfenidone ($n = 18$) and nintedanib ($n = 13$). The survival rate 90 days after AE-IPF onset was 61.1 % in the pirfenidone group and 61.5 % in the nintedanib group.

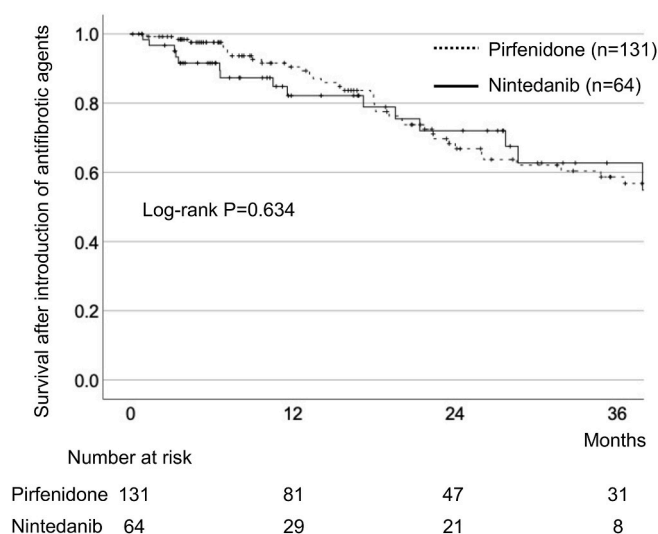


Fig. 5. Kaplan–Meier survival curves after introduction of antifibrotic agents. Patients treated with pirfenidone ($n = 131$) and nintedanib ($n = 64$) had similar survival curves (log rank test, $P = 0.634$).

capacity (FVC) decline as compared with placebo. In INPULSIS-2, 3.6 % of patients treated with nintedanib developed AE-IPF after 1 year, as compared with 9.6 % per year in placebo-treated patients. In contrast, a nintedanib group and placebo group had similar AE-IPF incidences in INPULSIS-1 (6.1 % vs. 5.4 %). Investigator-reported incidence of AE-IPF was lower in a nintedanib treatment group than a placebo group in pooled analysis [16]. Moreover, integrated analysis of the TOMORROW and INPULSIS trials showed lower AE-IPF risk in a nintedanib treatment group [15]. In the present nintedanib treatment group, AE incidence was higher than in clinical trials, perhaps because our patients had more-severe disease at baseline. Lower baseline FVC was a risk factor for AE-IPF in pooled analysis of INPULSIS [16]. Baseline percent predicted FVC (%FVC) was 79.5 ± 17.0 % in nintedanib-treated patients in INPULSIS-1 and 80.1 ± 18.1 % in INPULSIS-2. Mean %FVC was 65.6 ± 19.7 %, which indicates that our patients had relatively severe restrictive impairment and lower diffusing capacity. Ethnic differences may also explain the discrepancy between past and present findings. AE-IPF incidence varies in relation to ethnicity, and Japanese patients may be more susceptible to AE-IPF [21]. In a subgroup analysis of Japanese patients in INPULSIS, 12.0 % of the placebo group developed AE-IPF, as compared with 7.6 % of the overall population [22]. However, AE incidence was similarly reduced among Japanese and the overall population in the nintedanib treatment group. Future studies should attempt to identify the reasons for these discrepant results.

AE-IPF incidence was lower in the pirfenidone group than in the nintedanib group in this study. To our knowledge, no study has directly compared AE-IPF incidence in patients treated with pirfenidone and nintedanib in clinical settings. In light of findings from clinical trials that antifibrotic drugs might prevent AE-IPF development, our data suggest that pirfenidone might have a stronger effect on suppressing AE-IPF. The mechanism by which antifibrotic agents protect against AE-IPF is unknown. The delay in loss of lung function attributable to antifibrotic agents may make patients less likely to develop AE-IPF [18]. Thus, the strong antifibrotic effect may be associated with protection against AE-IPF. However, no clinical trial has directly compared the antifibrotic effects of these agents. Because of the small sample size, we could not compare FVC decline after the start of antifibrotic therapy in this study. Antifibrotic agents might also be effective for treatment of AE-IPF. In a retrospective analysis, patients had better survival after AE-IPF onset when pirfenidone was added to conventional treatment including high-dose corticosteroids [23,24]. Similarly, some case reports found that nintedanib was useful for treatment of AE-IPF [25,26].

Neutrophilic–lymphocytic inflammation, subsequent increases in transforming growth factor- β activity, and capillary injury have been implicated in the pathogenesis of AE-IPF [27,28]. Therefore, authors speculated that the anti-neutrophilic, anti-lymphocytic inflammatory effect, normalizing permeability, and inhibiting growth factors of these drugs may be useful for treatment of AE-IPF. These therapeutic effects of antifibrotic agents might also act as suppressing effect for AE-IPF development.

The prognosis for AE-IPF is poor: median survival is 3–4 months [4, 5]. Although antifibrotic agents might have beneficial effect for treatment of AE-IPF [23–26], It is unclear whether these agents should be withheld or continued during AE-IPF [18]. In the current study, patients continued antifibrotic therapy during treatment for AE-IPF. Survival after the onset of AE-IPF might be better than the previous studies [4,5], suggesting the positive effect of antifibrotic agents against AE-IPF in acute phase. Severity at AE-IPF onset and outcome after AE-IPF were similar in the pirfenidone and nintedanib groups. Clarifying differences between these drugs in therapeutic efficacy for AE-IPF will require additional research.

Some putative risk factors for AE-IPF have been identified. Reduced pulmonary function, never-smoking status, and high baseline KL-6 concentration predicted development of AE-IPF [4,16,17,24]. Although we observed similar tendencies, we found no significant differences in these factors between groups, perhaps because of the relatively small sample size of this study and administration of antifibrotic agents. Unlike other studies, the present study was restricted to patients using antifibrotic agents.

This study has several limitations. First, it was a retrospective study conducted at a single center. However, diagnosis and treatment of the present patients were consistent. Second, some patients received antifibrotic agents after AE-IPF development. However, the proportions who developed AE-IPF before the start of antifibrotic therapy were similar in the pirfenidone group and nintedanib group. That explains, in part, why a relatively high proportion of patients received corticosteroids in combination with antifibrotic agents, regardless of their contraindication for IPF [1]. Finally, our study lacked a control group of patients not treated with antifibrotic agents. Because clinical trials support the efficacy of these antifibrotic agents and international guidelines recommend their use for IPF treatment, we were ethically required to provide antifibrotic therapy to our patients. In addition, patients with mild disease were generally only observed. If their condition worsened, drug treatment was selected in accordance with disease behavior. Thus, both clinical characteristics and disease behavior differed greatly between the treatment and non-treatment groups. Therefore, in this retrospective study we could not compare the present patients with those who did not receive antifibrotic agents. This comparison should be attempted in a future clinical trial.

5. Conclusions

We retrospectively analyzed AE-IPF incidence in patients receiving antifibrotic agents. Our real-world experience suggests that pirfenidone has a greater effect than nintedanib on suppressing AE-IPF incidence. Although these data deepen our understanding of IPF treatment, a large-scale study is necessary to confirm the present results.

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CRedit authorship contribution statement

Takuma Isshiki: had access to the data and assumes responsibility for their accuracy, All authors were involved in drafting and revising the

manuscript and gave their final approval of the version to be published, contributed to collection of the data, All authors were involved in drafting and revising the manuscript and gave their final approval of the version to be published. **Susumu Sakamoto**: contributed to the design of the study, All authors were involved in drafting and revising the manuscript and gave their final approval of the version to be published. **Akira Yamasaki**: contributed to collection of the data, All authors were involved in drafting and revising the manuscript and gave their final approval of the version to be published. **Hiroshige Shimizu**: contributed to collection of the data, All authors were involved in drafting and revising the manuscript and gave their final approval of the version to be published. **Shion Miyoshi**: contributed to collection of the data, All authors were involved in drafting and revising the manuscript and gave their final approval of the version to be published. **Yasuhiko Nakamura**: contributed to collection of the data, All authors were involved in drafting and revising the manuscript and gave their final approval of the version to be published. **Sakae Homma**: contributed to the design of the study, All authors were involved in drafting and revising the manuscript and gave their final approval of the version to be published. **Kazuma Kishi**: contributed to the design of the study, All authors were involved in drafting and revising the manuscript and gave their final approval of the version to be published.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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